

PANCREATIC SECRETIONS

COMPOSITION AND

FUNCTION

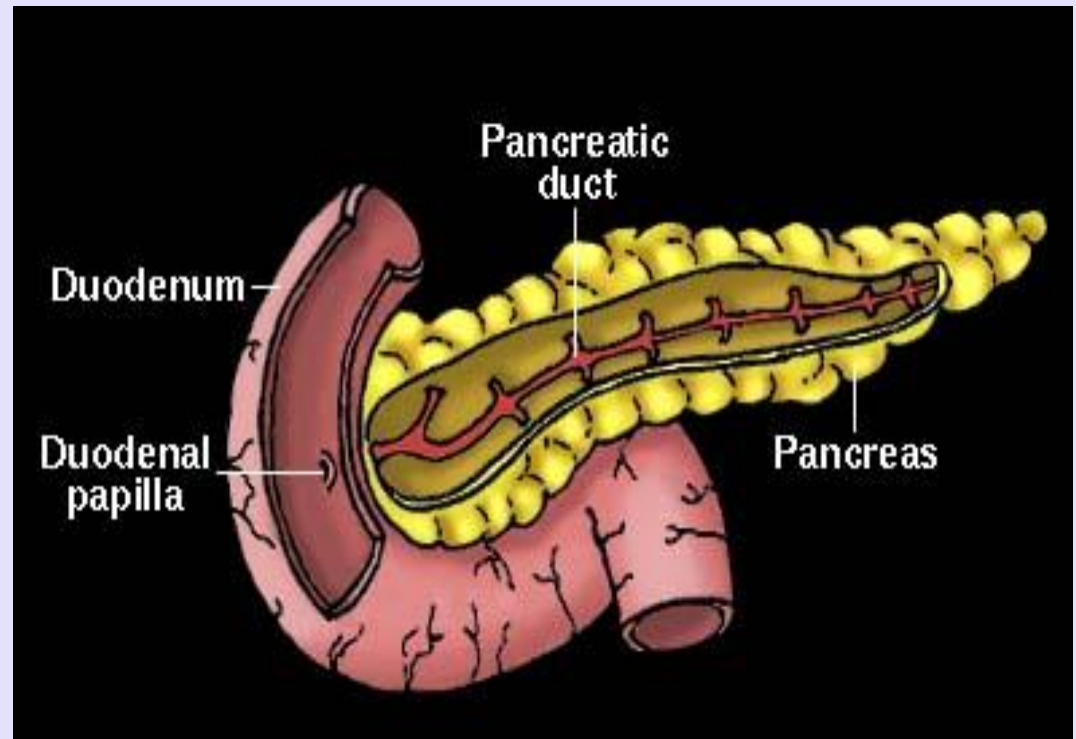
LEARNING *OBJECTIVES*

At the end of lecture students should be able to

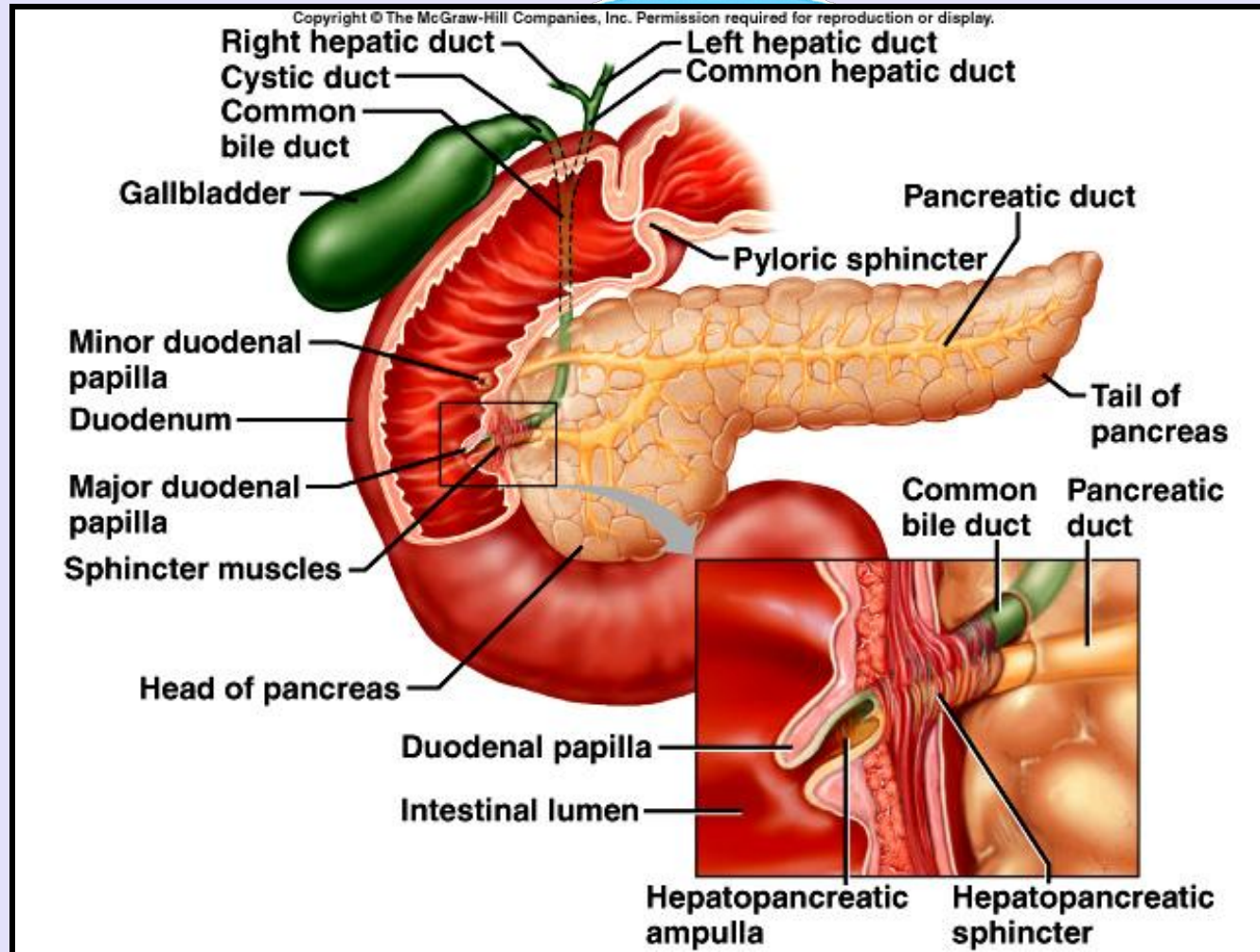
- Describe the structure of pancreas
- List the secretions of pancreas, their cells of origin & their functions
- Differentiate b/w exocrine & endocrine pancreas
- Discuss the regulation of pancreatic secretion
- Illustrate the different phases of pancreatic secretions

SECRETION OF THE PANCREAS

- Endocrine - insulin & glucagon
- Exocrine - enzymes and bicarbonate
- Essential for digestion
- Almost under separate hormonal control

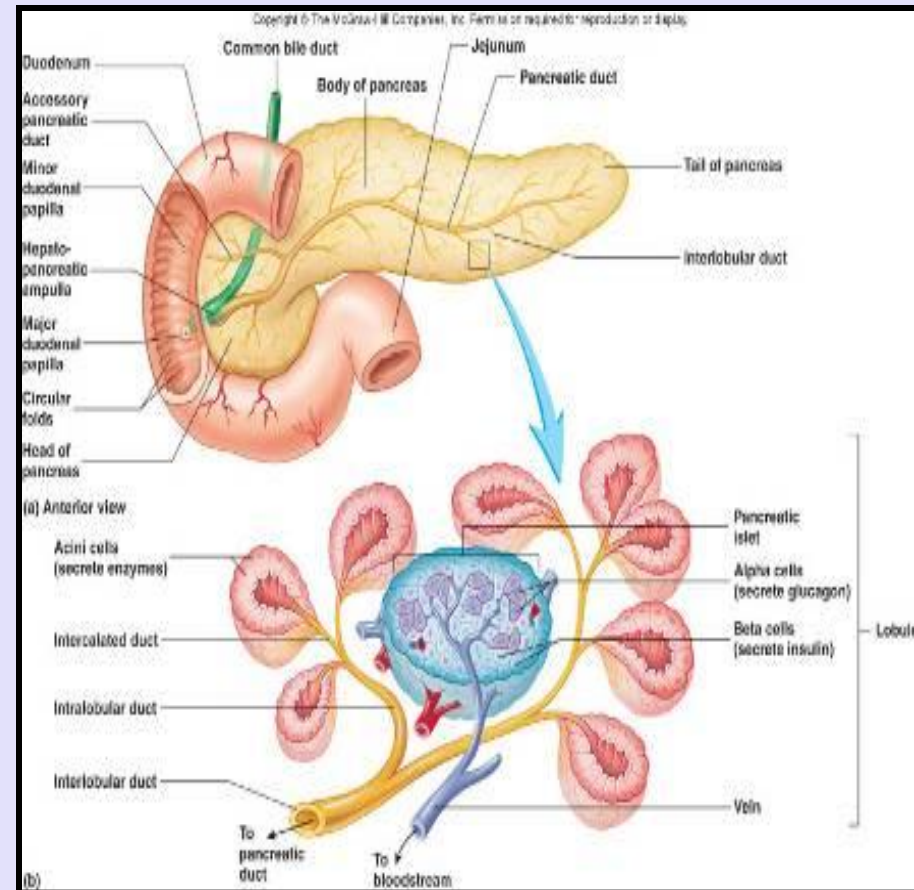


PANCREAS

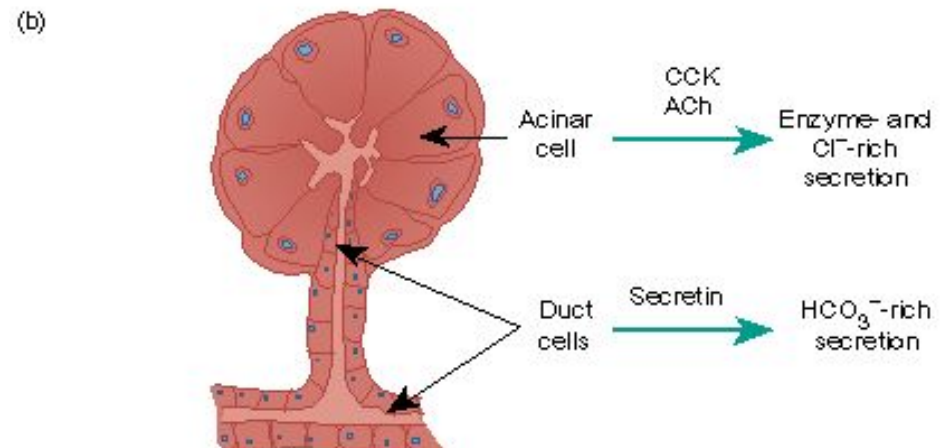
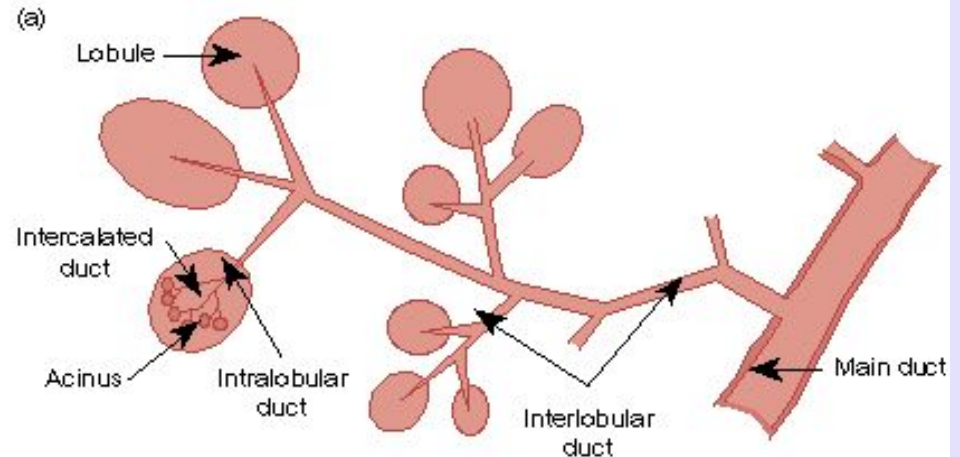
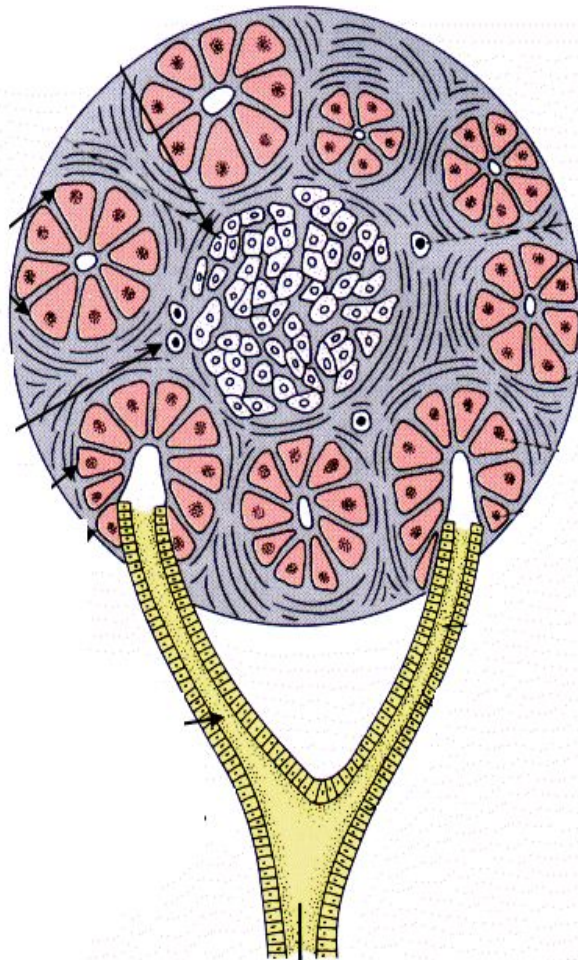


PANCREAS

- **Pancreas has both endocrine and exocrine parts**
- **Endocrine: pancreatic islets**
Produce insulin, glucagon, and somatostatin
- **Exocrine: groups acini (grape-like cluster) form lobules separated by septa.**
- **Intercalated ducts lead to interlobular ducts which leads to the pancreatic duct**
- **Pancreatic duct joins common bile duct and enters duodenum at the hepato pancreatic ampulla controlled by the hepatopancreatic ampullar sphincter**



ANATOMY AND SECRETION

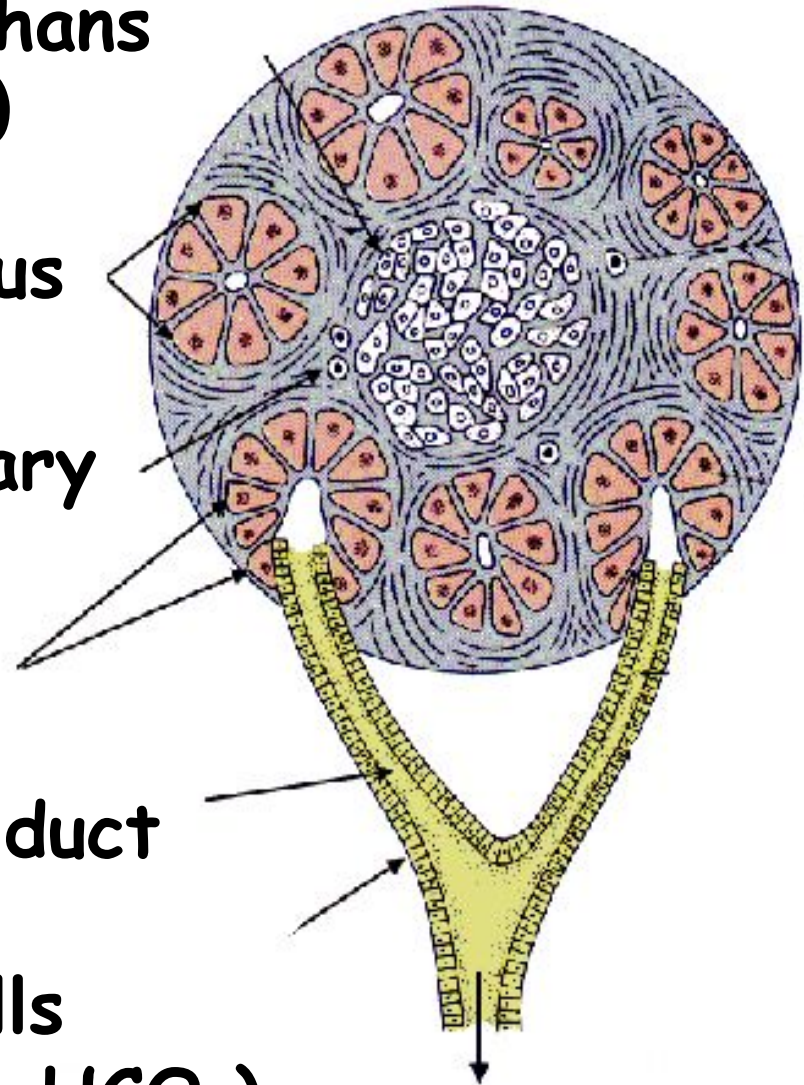


Islet of Langerhans (secrete insulin)

Acinus
Capillary
Acinar cells
(secrete enzymes)
Intercalated duct

Duct cells
(secrete HCO_3^-)

To pancreatic duct



EXOCRINE PANCREAS

Responsible for digestive function of pancreas

- **Anatomical Structure**

Acini → Ducts → Pancreatic Duct

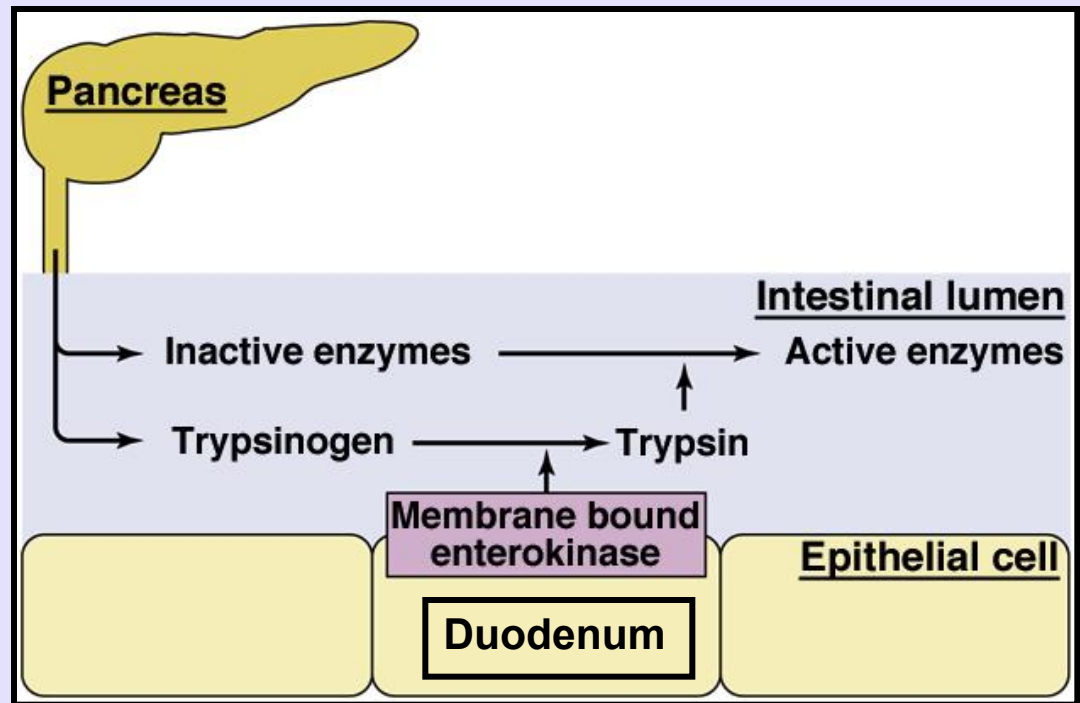
- **Function**

Secretion of bicarbonate by duct cells

Secretion of digestive enzymes by acinar cells

ZYMOGENS

- Acinar cells contain digestive enzymes stored as inactive zymogen granules
- Prevents autodigestion of pancreas
- Enterokinase (bound to brush border of duodenal enterocytes) converts trypsinogen to trypsin
- Trypsin converts all other zymogens to active forms



CATEGORIES OF PANCREATIC ENZYMES

Proteases	Cleave peptide bonds
Nucleases	Hydrolyze DNA/RNA
Elastases	Collagen digestion
Phospholipases	Phospholipids to fatty acids
Lipases	Triglycerides to fatty acids+ glycerol
Amylase	Starch to maltose + glucose

PANCREATIC SECRETIONS: PANCREATIC JUICE

- Aqueous. Produced by columnar epithelium lining smaller ducts. Na^+ , K^+ , HCO_3^- , water. Bicarbonate lowers pH inhibiting pepsin and providing proper pH for enzymes

Enzymatic portion.

- trypsinogen,
- chymotrypsinogen,
- procarboxypeptidase,
- pancreatic amylase,
- pancreatic lipase,
- deoxyribonuclease and ribonuclease.

PANCREATIC SECRETIONS: PANCREATIC JUICE

- Interaction of duodenal and pancreatic enzymes.
 - Enterokinase from the duodenal mucosa and attached to the brush border activates trypsinogen to trypsin.
 - Trypsin activates chymotrypsinogen to chymotrypsin.
 - Trypsin activates procarboxypeptidase to carboxypeptidase.
- Trypsin, chymotrypsin and carboxypeptidase digest proteins: proteolytic.
- Pancreatic amylase continues digestion of starch
- Pancreatic lipase digests lipids
- Deoxyribonucleases and ribonucleases digest DNA and ribonucleic acid, respectively

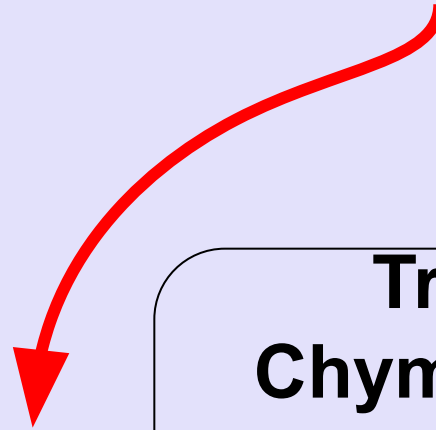
ACTIVATION OF PANCREATIC PROTEASES

Enterokinase

Trypsinogen



Trypsin



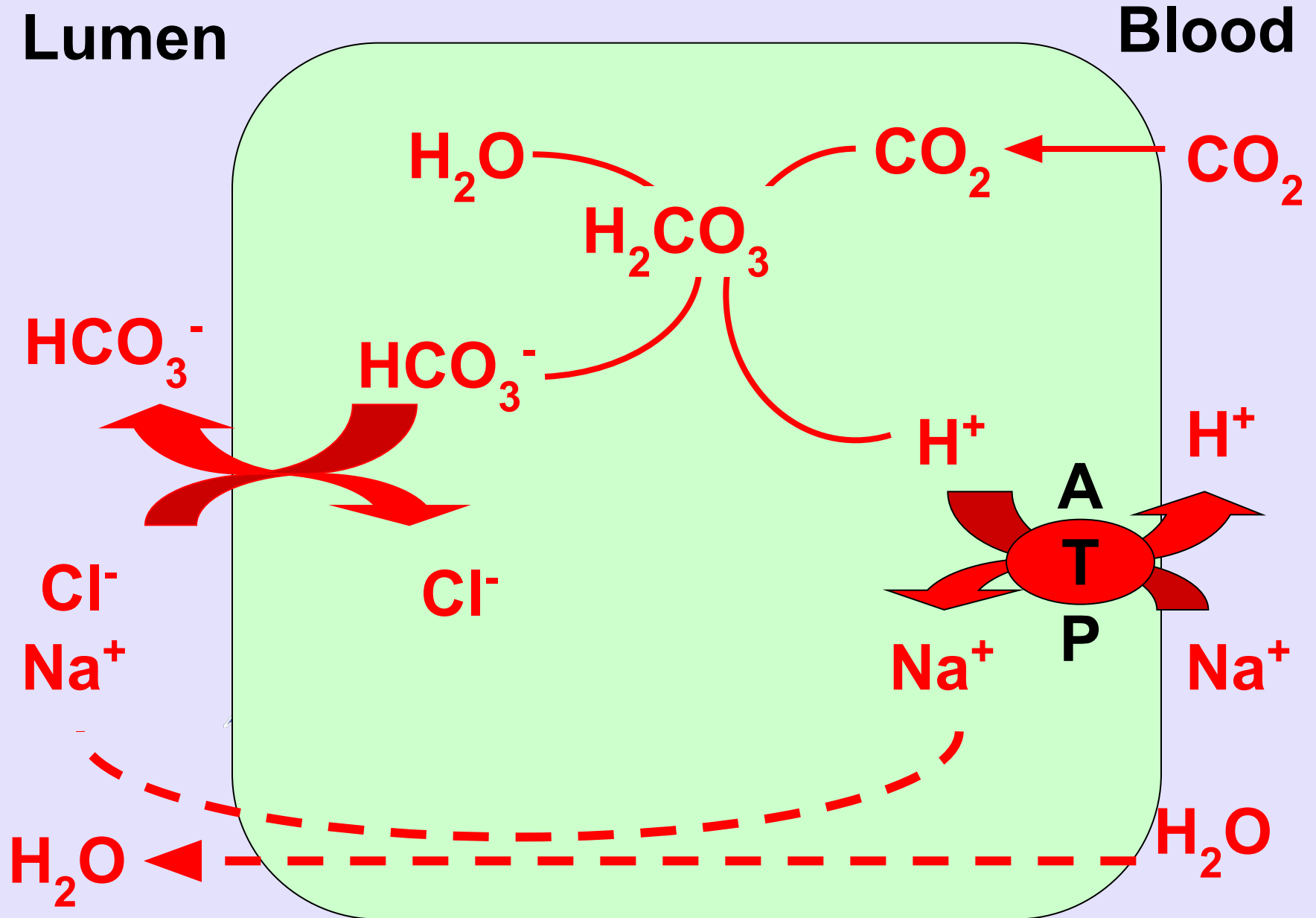
**Trypsinogen
Chymotrypsinogen
Proelastase
Procarboxypeptidase**



**Trypsin
Chymotrypsin
Elastase
Carboxypeptidase**

- Same pancreatic cells which secrete the proteolytic enzymes, also secrete the trypsin inhibitors, that inhibit TRYPSIN otherwise the normal cells of pancreas would have been digested by trypsin
- When pancreas is damaged or pancreatic duct is blocked, large quantities of pancreatic secretions are pooled and enzymes becomes activated which digest the pancreas within hours(**acute pancreatitis**---leads to shock)

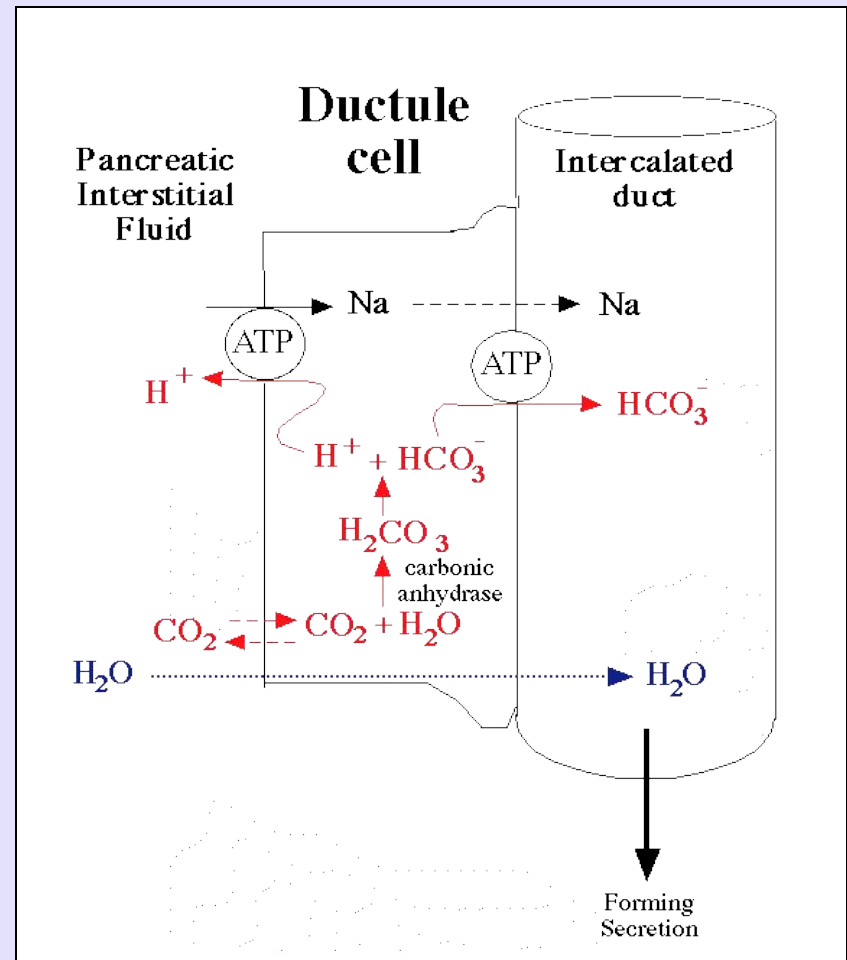
BICARBONATE SECRETION



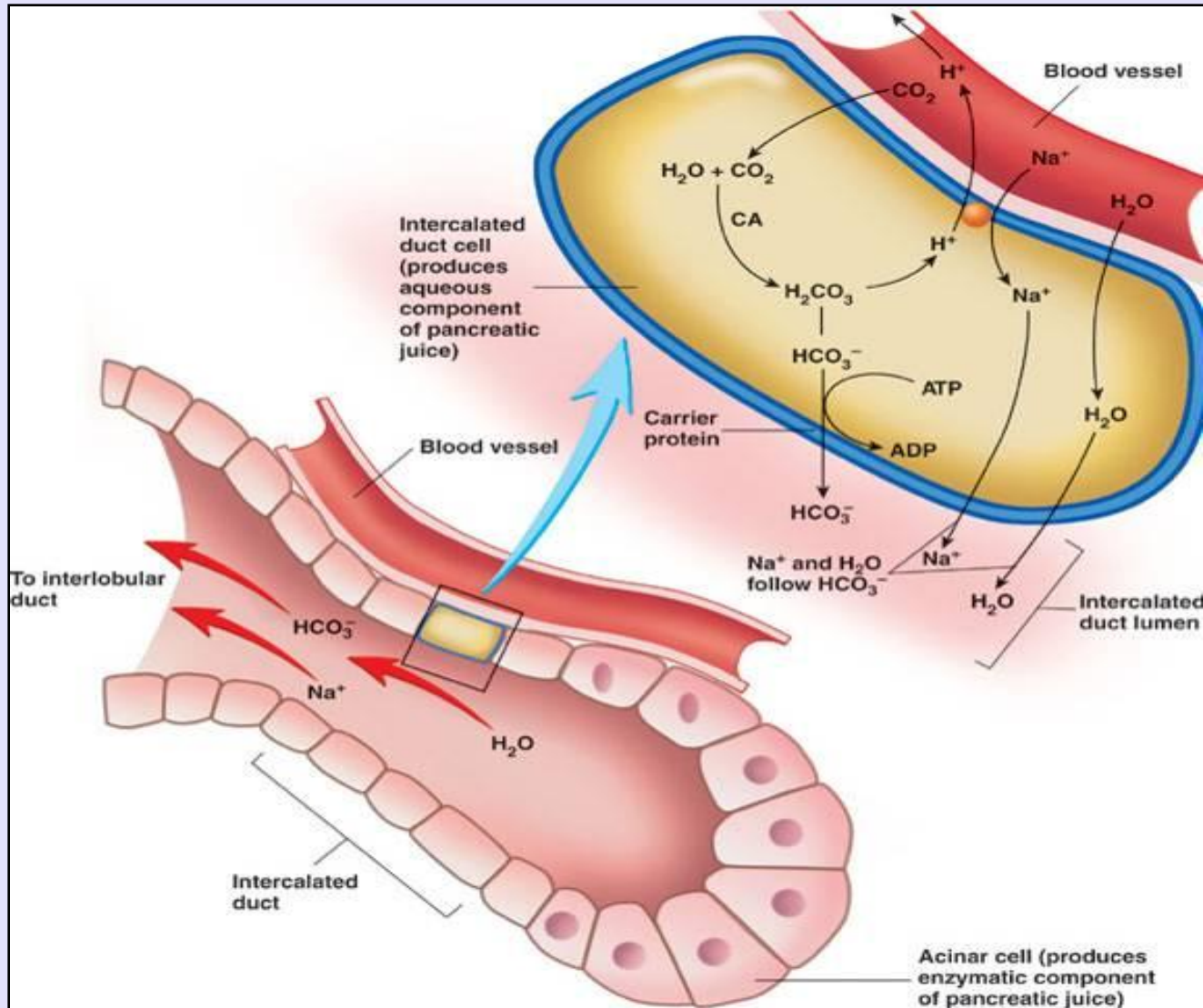
BICARBONATE FUNCTION

Functions

1. Neutralize gastric acid emptied into the duodenum
2. Provide a favorable alkaline environment for optimal activity of pancreatic enzymes



BICARBONATE ION PRODUCTION IN PANCREAS



CONTROL OF PANCREATIC FUNCTION

- **Bicarbonate secretion stimulated by secretin**
- **Secretin released in response to acid in duodenum**
- **Zymogen secretion stimulated by cholecystokinin (CCK)**
- **CCK released in response to fat/amino acids in duodenum**
- **Also under neural control (vagal/local reflexes) - triggered by arrival of organic nutrients in duodenum**

CONTROL OF PANCREATIC SECRETION

- secretion in 3 phases:

- Cephalic phase - only 10-15% of total secretion
- Activation of vagal efferent stimulates enzyme release
- Gastric phase - only present in some species
- **NOT SIGNIFICANT IN HUMANS**
- Intestinal phase - majority of secretion
- Combination of hormones CCK and secretin and neuronal reflex
- Results in maximal enzyme and bicarbonate release

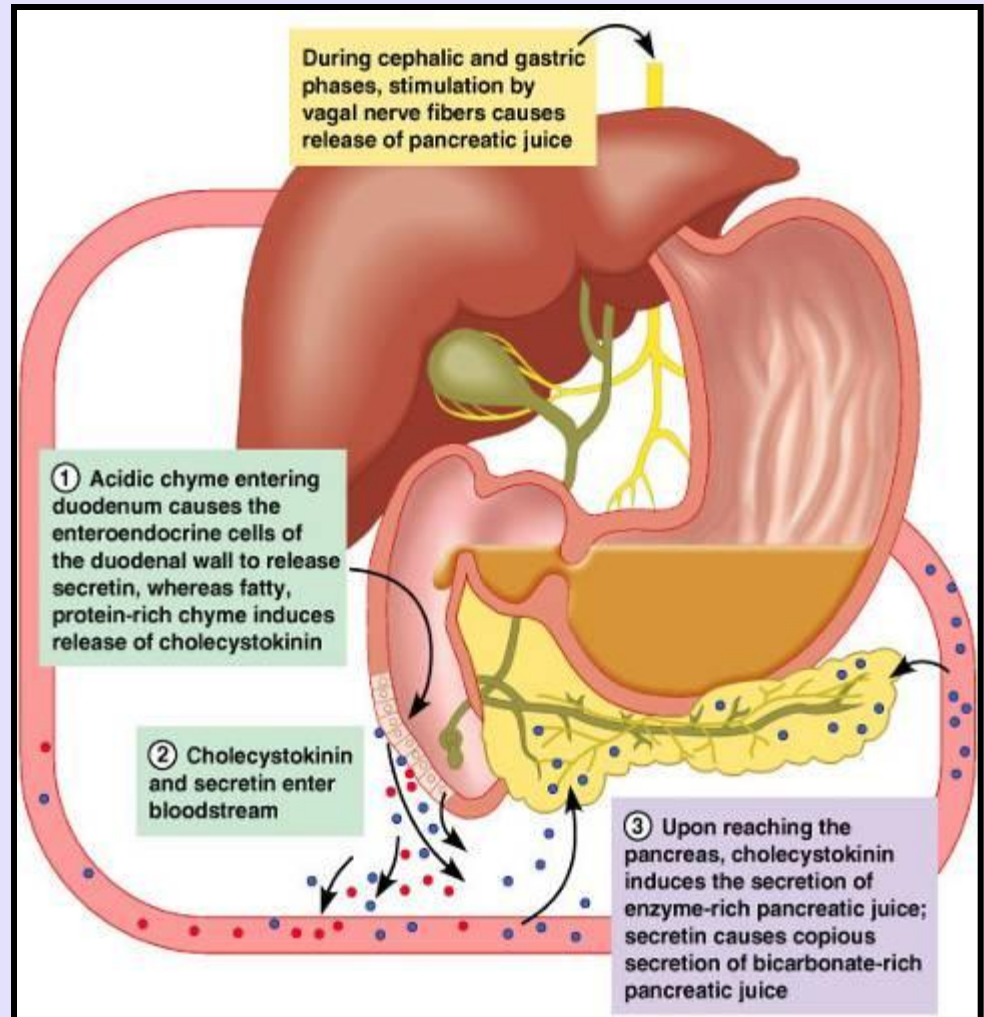
CONTROL OF PANCREATIC SECRETION

Key hormones in stimulation of secretion are:

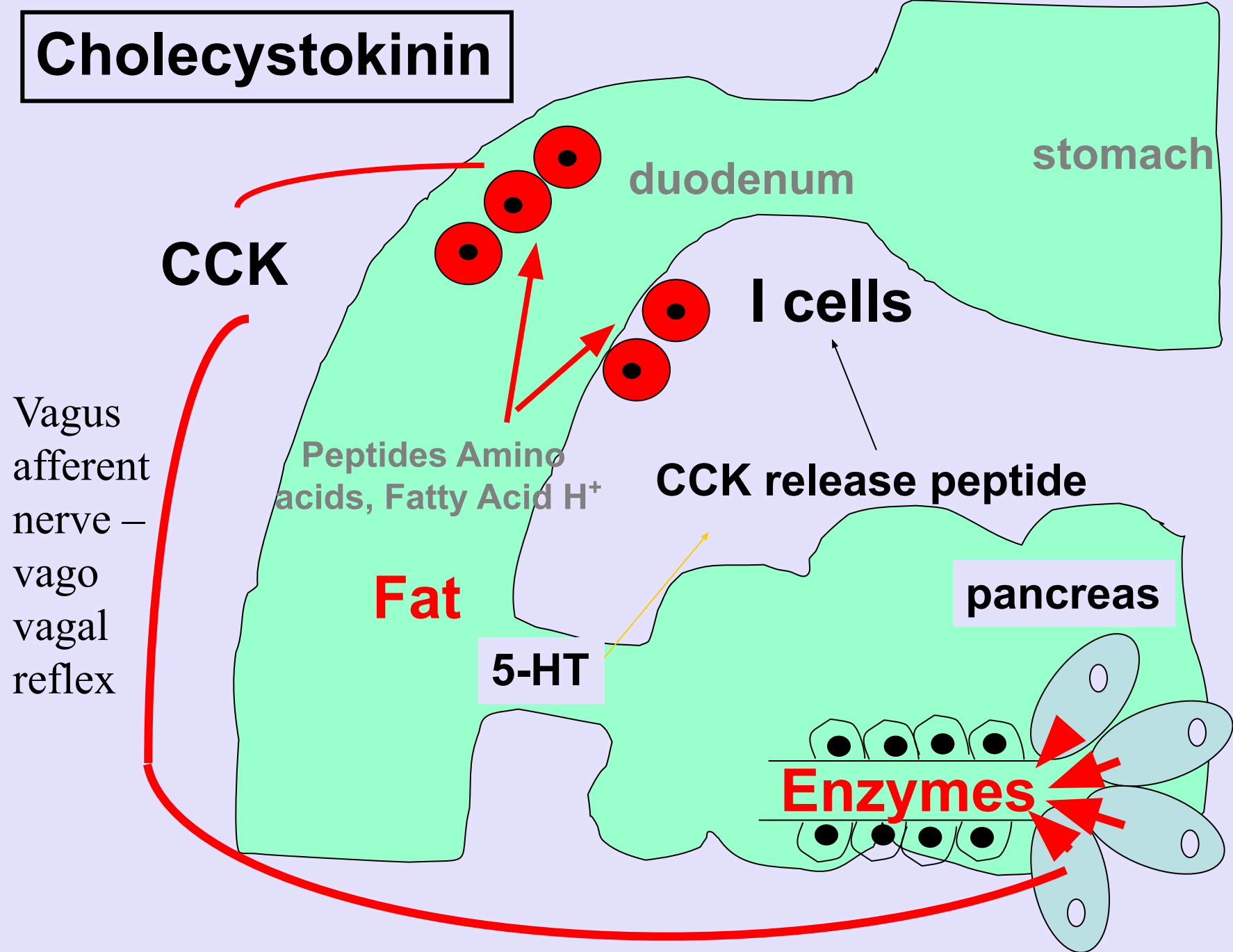
- **Cholecystokinin (CCK)** and **Secretin** (released from the small intestine)
- **Inhibiting factors:** SS, PP, glucagon

REGULATION OF PANCREATIC SECRETION

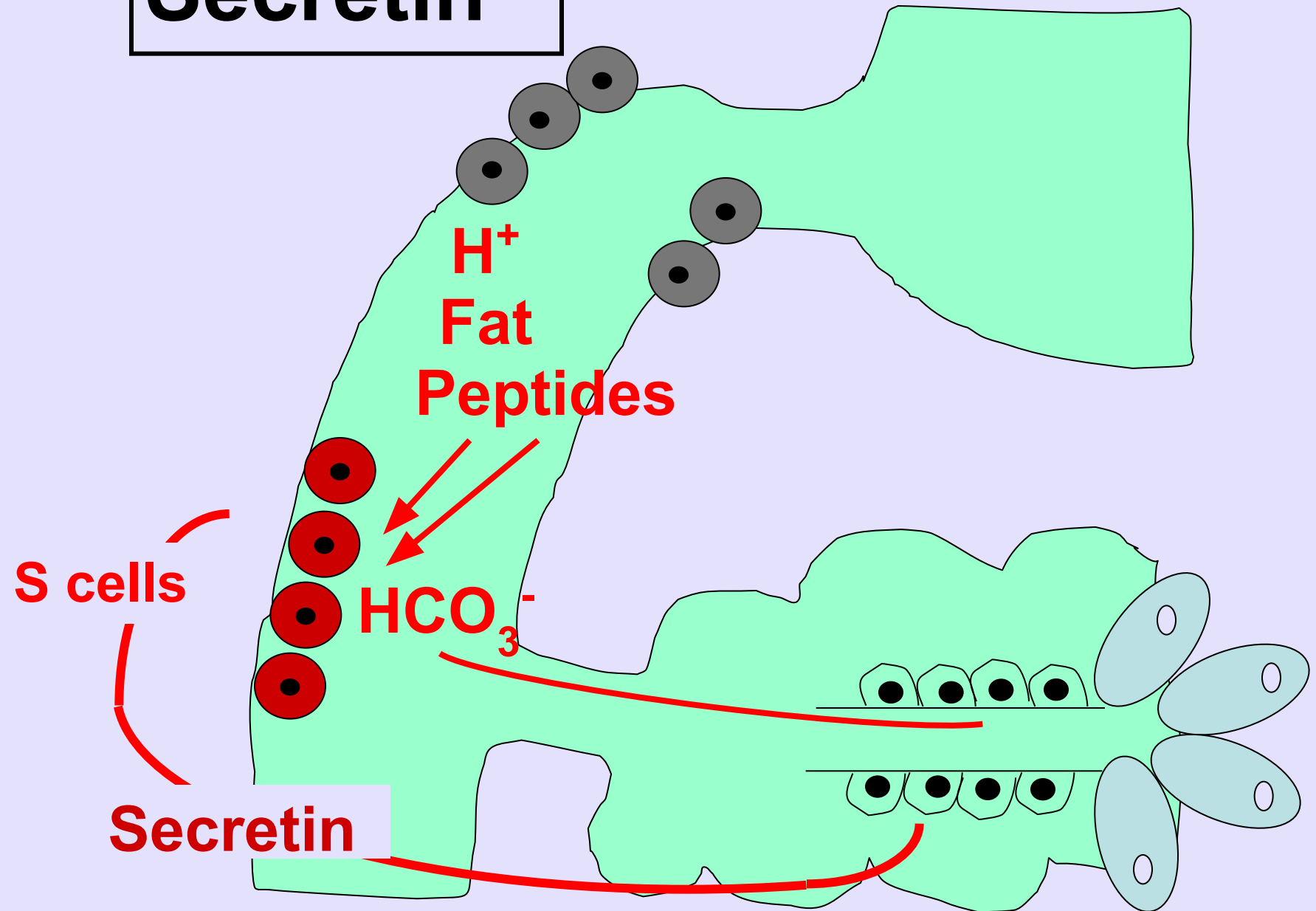
- Stimuli:
 - 1 **Acetyl choline**
(vagus n. ending)
 - 2 **CCK**
 - 3 **Secretin**



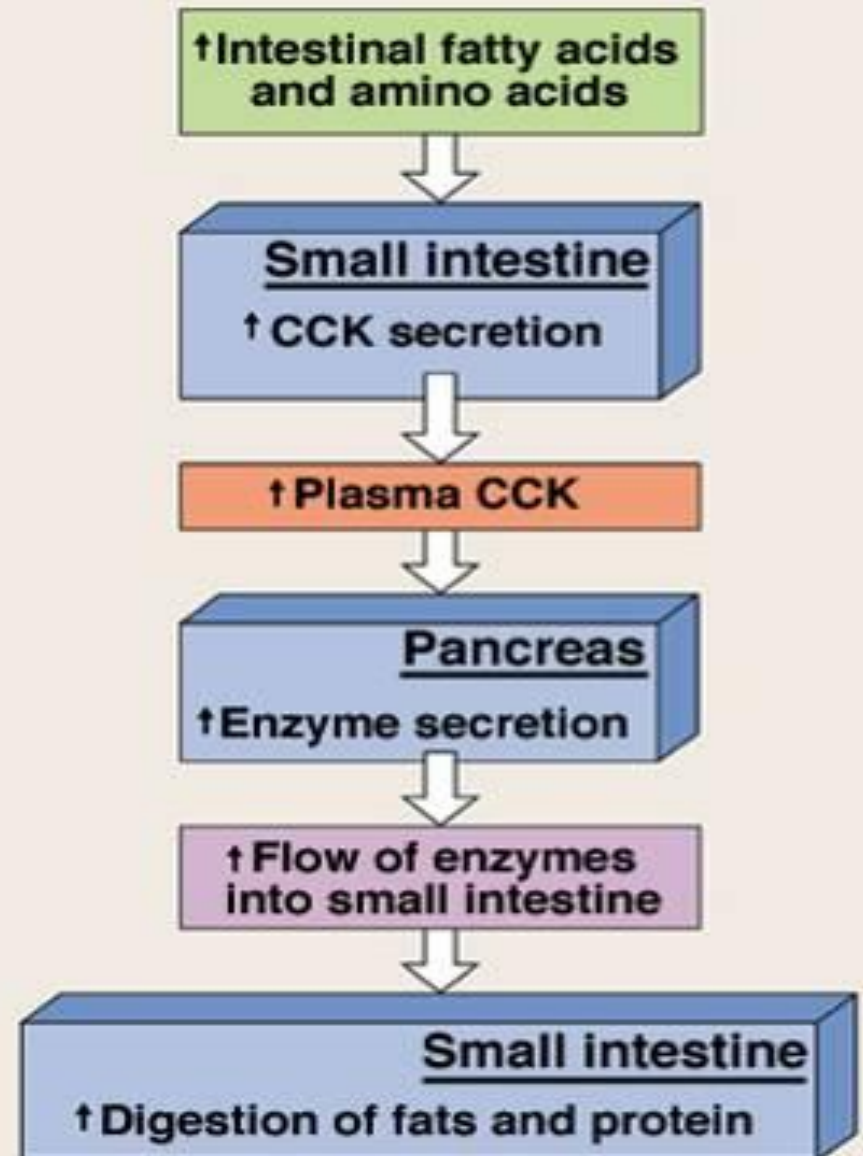
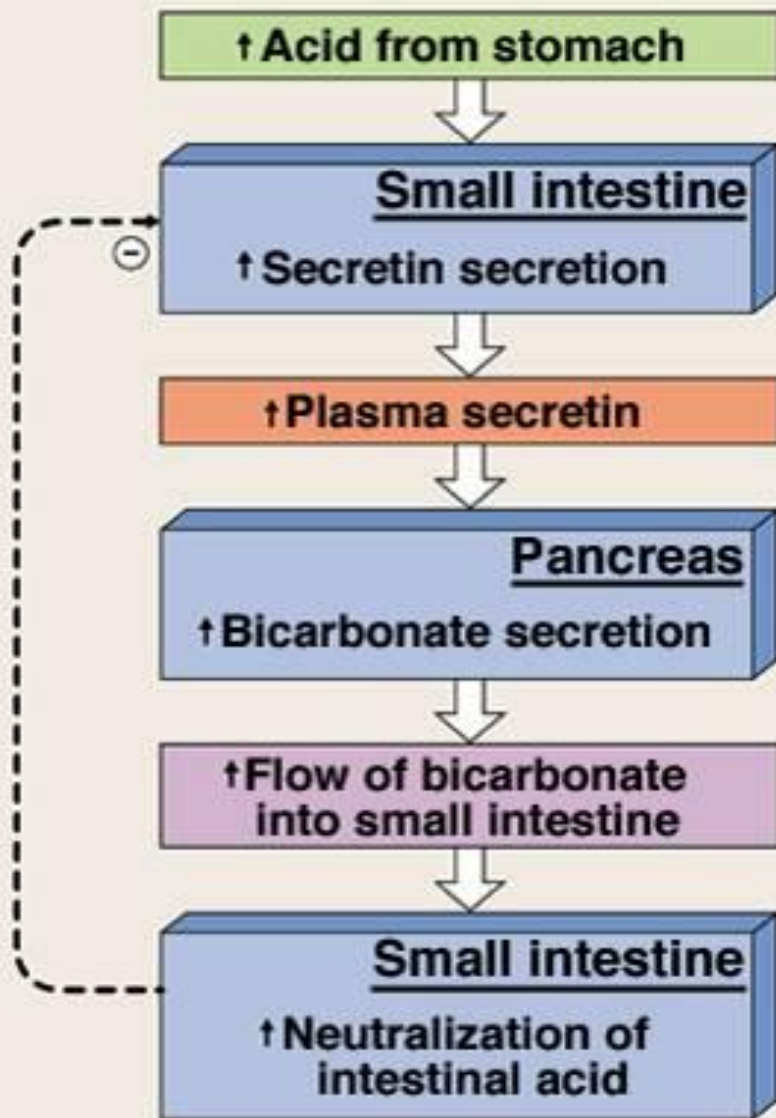
Cholecystokinin



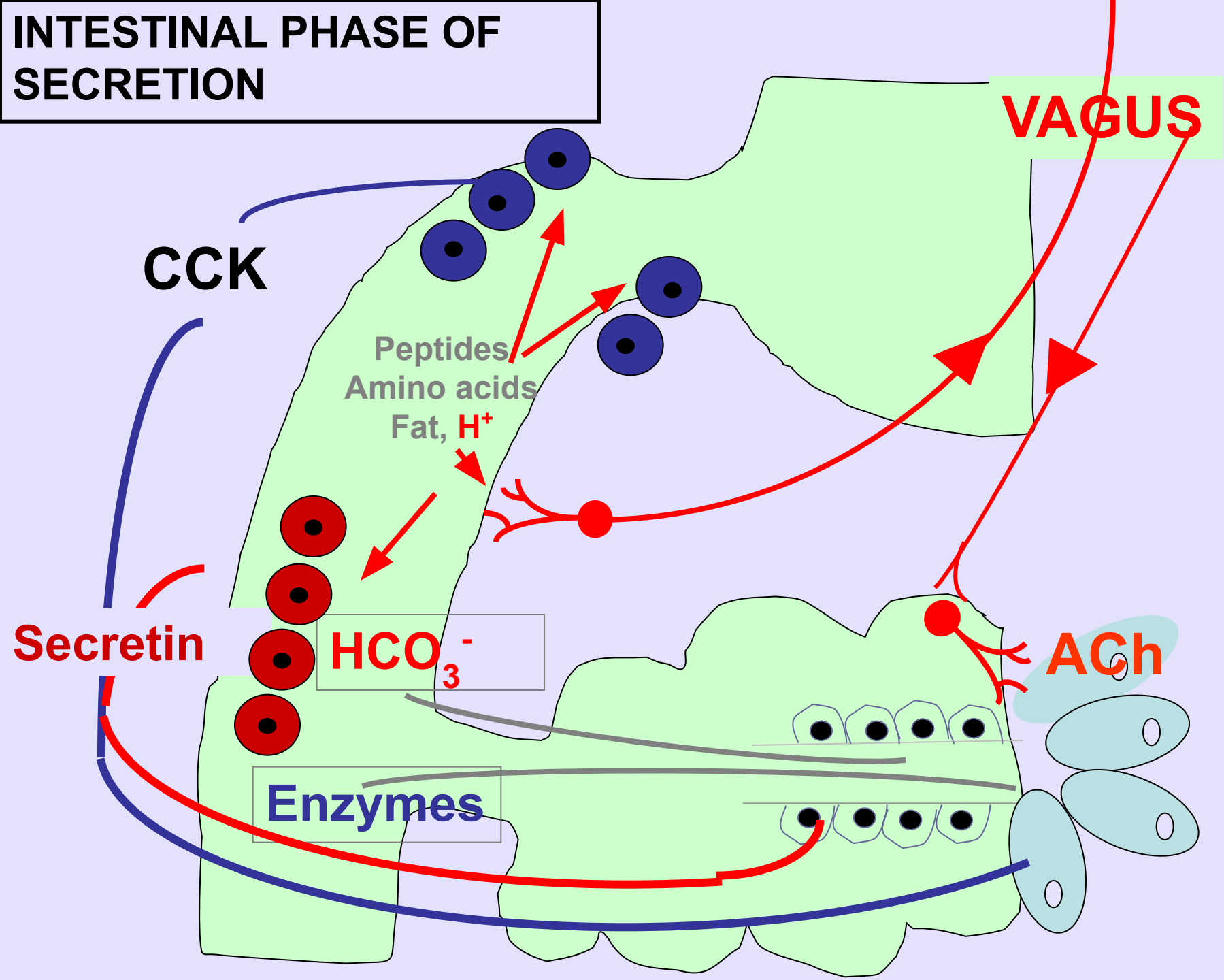
Secretin



CONTROL OF PANCREATIC FUNCTION



INTESTINAL PHASE OF SECRETION



THANK YOU